

Notes on method of fundamental solutions for solving Laplace equation

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This note is to discuss a method of fundamental solutions (MFS) for solving Laplace equation, based on the recent work by Broms et al.¹. The system we are interested in is the dielectric spheres excited by a constant electric field.

I. PROBLEM SETUP

The background medium has a dielectric constant $\epsilon_0 = 1$, and the spheres have dielectric constant ϵ_r . Without loss of generality, we assume the electric field is along z -direction, denoted as E_0 . The resulting potential $\phi(x)$ satisfies the Laplace equation

$$\begin{cases} \Delta_x \phi(x) = 0, & x \in \Omega_0, \Omega_i \\ \phi^+(x)|_{\partial\Omega_i} = \phi^-(x)|_{\partial\Omega_i}, & x \in \partial\Omega_i \\ \epsilon_0 \partial_n \phi^+(x)|_{\partial\Omega_i} = \epsilon_r \partial_n \phi^-(x)|_{\partial\Omega_i}, & x \in \partial\Omega_i \\ \phi(x) \rightarrow -E_0 z, & |x| \rightarrow \infty \end{cases} \quad (1)$$

where Ω_0 is the background domain, Ω_i is the domain of the i -th sphere, $\partial\Omega_i$ is the boundary of the i -th sphere, and n is the normal direction pointing from Ω_i to Ω_0 .

We then split the potential into two parts, $\phi(x) = u_{inc}(x) + u(x)$, where $u_{inc}(x) = -E_0 z$ is the incident potential, and $u(x)$ is the scattered potential. The scattered potential then satisfies the following equations:

$$\begin{cases} \Delta_x u(x) = 0, & x \in \Omega_0, \Omega_i \\ u^+(x)|_{\partial\Omega_i} = u^-(x)|_{\partial\Omega_i}, & x \in \partial\Omega_i \\ \epsilon_0 \partial_n u^+(x)|_{\partial\Omega_i} - \epsilon_0 E_0 n_z = \epsilon_r \partial_n u^-(x)|_{\partial\Omega_i} - \epsilon_r E_0 n_z, & x \in \partial\Omega_i \\ u(x) \rightarrow 0, & |x| \rightarrow \infty \end{cases} \quad (2)$$

where n_z is the z -component of the normal vector.

II. SINGLE SPHERE

We first derive the formula for a single polarizable sphere. For a sphere with radius a centered at the origin of the coordinate system excited by a constant electric field E_0 along z -direction, the potential outside and inside the sphere can be solved analytically as

$$\phi(x) = \begin{cases} -E_0 z + \frac{a^3(\epsilon_r - 1)}{(\epsilon_r + 2)} \frac{E_0 z}{|x|^3}, & |x| > a \\ -\frac{3}{\epsilon_r + 2} E_0 z, & |x| < a \end{cases} \quad (3)$$

To solve this problem numerically, we make use of the method of fundamental solutions (MFS). For this problem, instead of use $\sigma(x)$ on the surface of the sphere, we place $p(x)$ and $q(x)$ inside and outside the sphere, respectively, which can contribute to the potential outside and inside the sphere, respectively. The idea is similar to the method of images, but the sources now located on sphere with radius $r_p < a$ and $r_q = a^2/r_p > a$. With $p(x)$ and $q(x)$, Eq. 2 can be rewritten as

$$\begin{cases} \int_{\Omega_p} G(x, y) p(y) dy - \int_{\Omega_q} G(x, y) q(y) dy = 0, & |x| = a \\ \int_{\Omega_p} \partial_n G(x, y) p(y) dy - \epsilon_r \int_{\Omega_q} \partial_n G(x, y) q(y) dy = -(\epsilon_r - 1) E_0 n_z, & |x| = a \end{cases} \quad (4)$$

Then to discretize the above equations, we use the t -design points on the sphere following Broms et al.¹, which discretize $p(x)$ and $q(x)$ into N equal weights points $\mathbf{p}$ and $\mathbf{q}$, and also discretize the boundary conditions on M points on the sphere surface, which are denoted as $\mathbf{n}_z$. The above equations can be rewritten in matrix form as

$$\begin{cases} \mathbf{S}_{pr} \mathbf{p} - \mathbf{S}_{qr} \mathbf{q} = \mathbf{0} \\ \mathbf{D}_{pr}^T \mathbf{p} - \epsilon_r \mathbf{D}_{qr}^T \mathbf{q} = -(\epsilon_r - 1) E_0 \mathbf{n}_z \end{cases} \quad (5)$$

where $\mathbf{S}$ and $\mathbf{D}^T$ are single and adjoint double layer potential matrices. Eq. (5) can be rewritten as follows:

$$\begin{bmatrix} \mathbf{S}_{pr} & \mathbf{S}_{qr} \\ \mathbf{D}_{pr}^T & \epsilon_r \mathbf{D}_{qr}^T \end{bmatrix} \begin{bmatrix} \mathbf{p} \\ -\mathbf{q} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ -(\epsilon_r - 1)E_0 \mathbf{n}_z \end{bmatrix} \quad (6)$$

which is a $2M \times 2N$ overdetermined linear system, which can be solved by least squares method.

III. FAR AWAY MULTIPLE SPHERES

We then consider the case of multiple spheres that are far away from each other.

A. Two spheres

For simplicity, we first consider the case of two spheres with the same radius a and dielectric constant ϵ_r , which are centered at $\mathbf{c}_1$ and $\mathbf{c}_2$, respectively. Again, we assume the electric field is along z -direction, denoted as E_0 , thus $u_{inc}(x) = -E_0 z$. The solution is again represented by MFS, where we place $p_i(x)$ and $q_i(x)$ inside and outside the i -th sphere, respectively, as shown in Fig. 1.

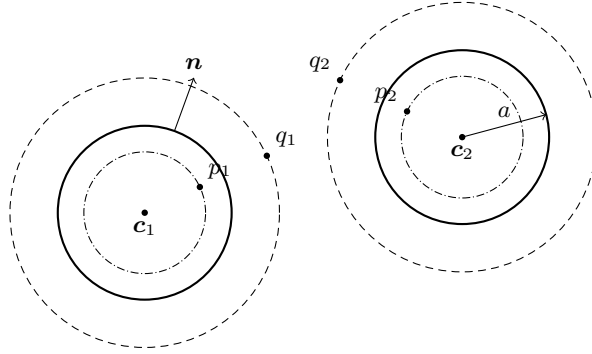


FIG. 1: Two-sphere setup (cross-section in the x - z plane) with MFS source points p_i (inside) and q_i (outside) for each sphere, centers $\mathbf{c}_i$, and applied field direction E_0 along $+z$.

The scattered potential can be represented as follows:

$$u(x) = \begin{cases} \int_{\Omega_{p_1}} G(x, y) p_1(y) dy + \int_{\Omega_{p_2}} G(x, y) p_2(y) dy, & x \in \Omega_0 \\ \int_{\Omega_{p_2}} G(x, y) p_2(y) dy + \int_{\Omega_{q_1}} G(x, y) q_1(y) dy, & x \in \Omega_1 \\ \int_{\Omega_{p_1}} G(x, y) p_1(y) dy + \int_{\Omega_{q_2}} G(x, y) q_2(y) dy, & x \in \Omega_2 \end{cases} \quad (7)$$

with which the boundary conditions can be represented as follows:

$$\begin{cases} \int_{\Omega_{p_1}} G(x, y) p_1(y) dy - \int_{\Omega_{q_1}} G(x, y) q_1(y) dy = 0, & x \in \partial\Omega_1 \\ \int_{\Omega_{p_2}} G(x, y) p_2(y) dy - \int_{\Omega_{q_2}} G(x, y) q_2(y) dy = 0, & x \in \partial\Omega_2 \\ \int_{\Omega_{p_1}} \partial_n G(x, y) p_1(y) dy + (1 - \epsilon_r) \int_{\Omega_{p_2}} \partial_n G(x, y) p_2(y) dy - \epsilon_r \int_{\Omega_{q_1}} \partial_n G(x, y) q_1(y) dy = -(\epsilon_r - 1)E_0 n_z, & x \in \partial\Omega_1 \\ (1 - \epsilon_r) \int_{\Omega_{p_1}} \partial_n G(x, y) p_1(y) dy + \int_{\Omega_{p_2}} \partial_n G(x, y) p_2(y) dy - \epsilon_r \int_{\Omega_{q_2}} \partial_n G(x, y) q_2(y) dy = -(\epsilon_r - 1)E_0 n_z, & x \in \partial\Omega_2 \end{cases} \quad (8)$$

Using the same discretization as in the single sphere case, we can rewrite the above equations in matrix form as follows:

$$\begin{bmatrix} \mathbf{S}_{p_1 r_1} & \mathbf{0} & \mathbf{S}_{q_1 r_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_{p_2 r_2} & \mathbf{0} & \mathbf{S}_{q_2 r_2} \\ \mathbf{D}_{p_1 r_1}^T & (1 - \epsilon_r) \mathbf{D}_{p_2 r_1}^T & \epsilon_r \mathbf{D}_{q_1 r_1}^T & \mathbf{0} \\ (1 - \epsilon_r) \mathbf{D}_{p_1 r_2}^T & \mathbf{D}_{p_2 r_2}^T & \mathbf{0} & \epsilon_r \mathbf{D}_{q_2 r_2}^T \end{bmatrix} \begin{bmatrix} \mathbf{p}_1 \\ \mathbf{p}_2 \\ -\mathbf{q}_1 \\ -\mathbf{q}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ -(\epsilon_r - 1)E_0 \mathbf{n}_z \\ -(\epsilon_r - 1)E_0 \mathbf{n}_z \end{bmatrix} \quad (9)$$

where $\mathbf{S}_{ab}$ and $\mathbf{D}_{ab}^T$ are the single and adjoint double layer potential matrices for sources at a and targets at b .

¹ A. Boms, A. H. Barnett, and A.-K. Tornberg, Journal of Computational Physics **523**, 113636 (2025).